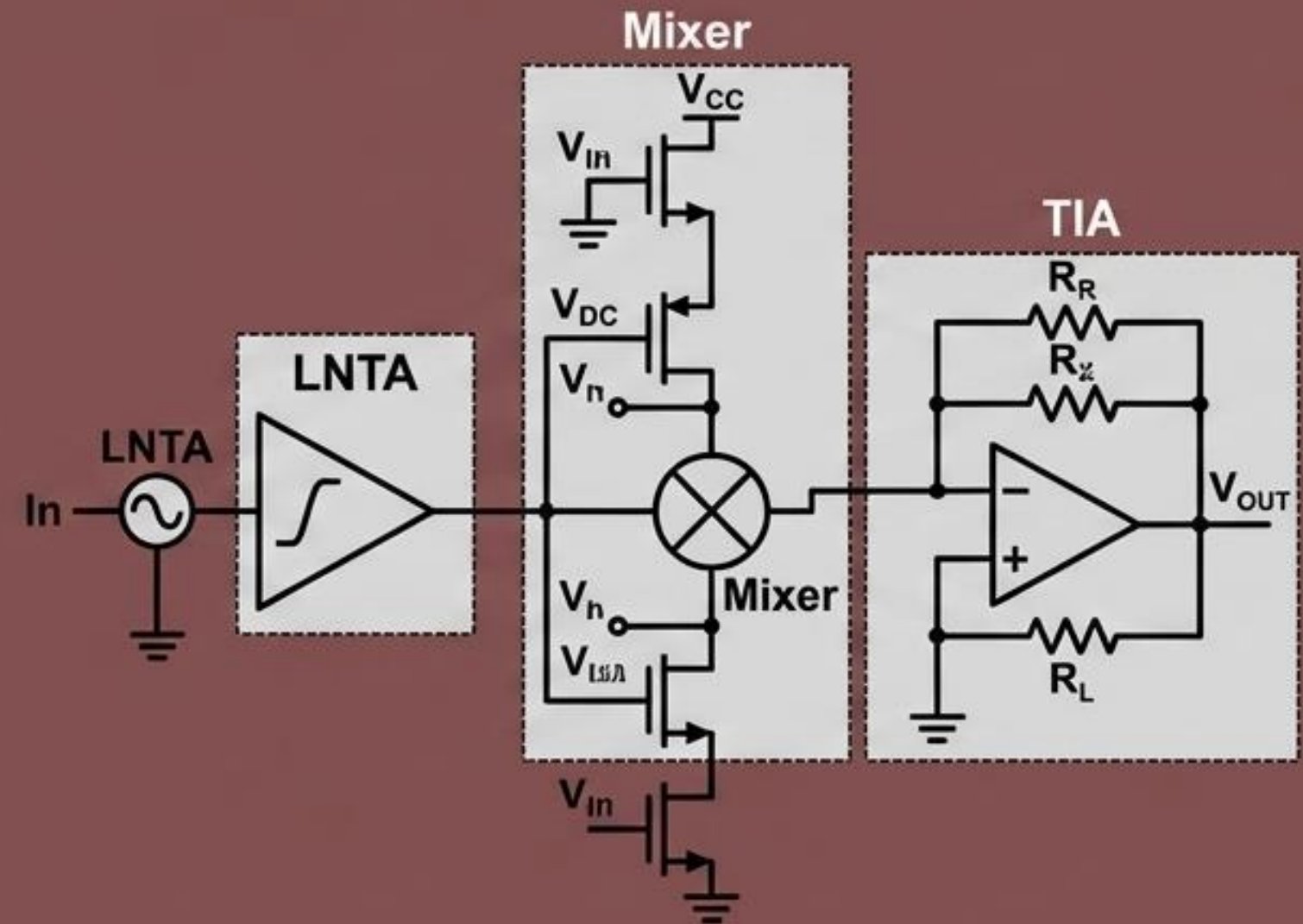


2.4 GHz Current-Mode Quadrature Receiver

Highly Linear, Low-Power RF
Front-End Design in 180nm CMOS

Optimized for Bluetooth Low
Energy (BLE) and Body Area
Networks (BAN)



Sri Harsha Darbhasayanam (UIN: 237007692),
Sheena Selvakumar (UIN: 537000654)
Texas A&M University, College Station

Team Member Contributions

Sri Harsha Darbhasayanam led the design, sizing, and performance optimization of the inductor-less Low Noise Transconductance Amplifier (LNTA), ensuring strict adherence to the noise figure, gain, and input matching requirements. He also contributed to the initial architecture of the passive mixer.

Sheena Selvakumar was primarily responsible for the development, optimization, and verification of the 25% duty-cycle 4-phase quadrature passive mixer, focusing on switching mechanics to minimize flicker noise and maximize conversion gain.

Both members collaborated equally on the integration of the RGC-TIA baseband stage, system-level transient simulations.

Voltage-Mode (Traditional Bottleneck)



Clipping and non-linearity due to limited voltage headroom.

Current-Mode (The Paradigm Shift)

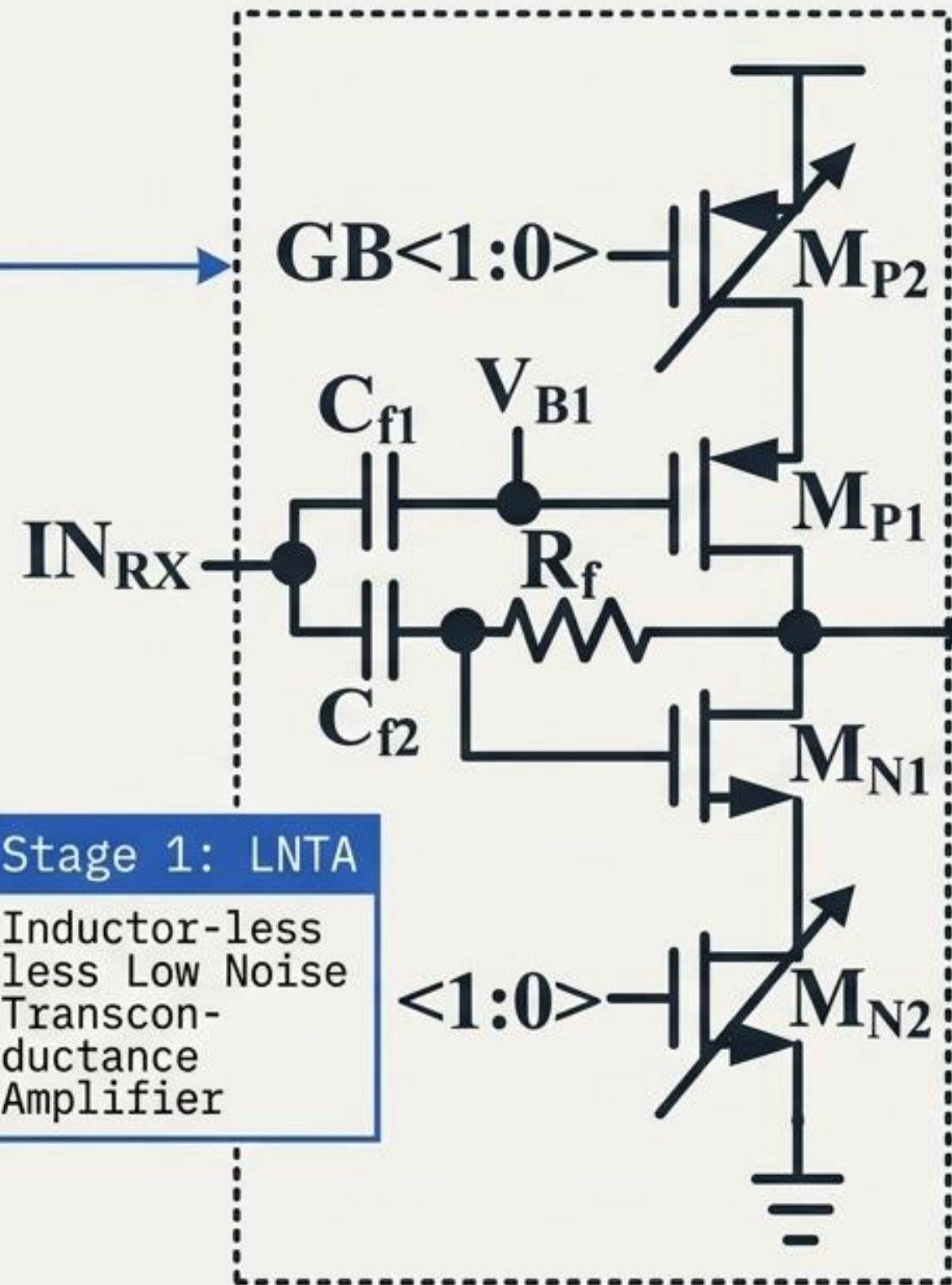


Signal flows unimpeded; decoupling linearity from voltage gain.

By converting to and maintaining the signal in the current domain through down-conversion, we completely decouple linearity from voltage gain.

 **The 3-Stage Solution:** LNTA -> Passive Mixer -> RGC-TIA

LNTA

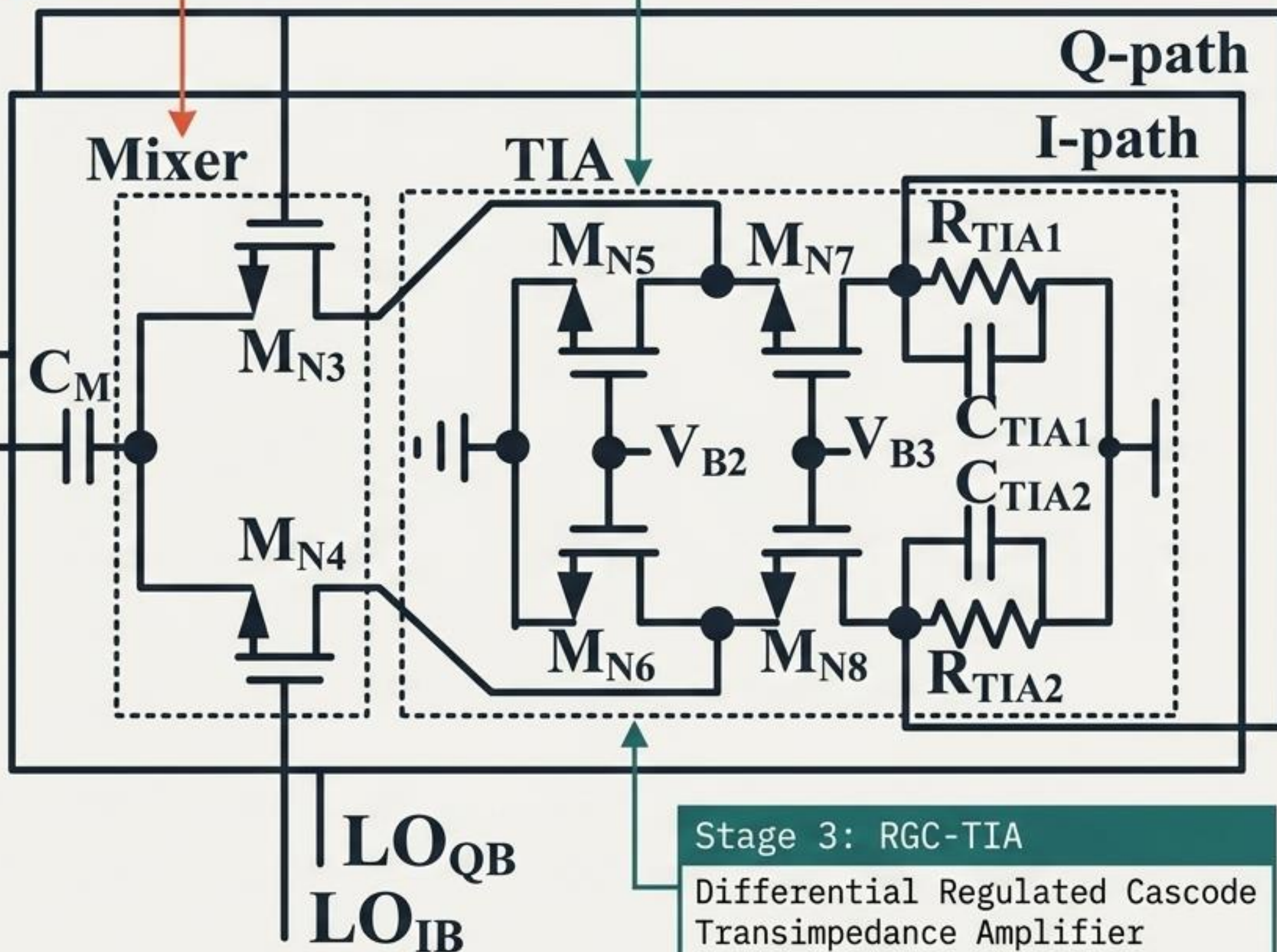


Stage 2: Passive Mixer

25% Duty-Cycle 4-Phase
Down-Conversion

Stage 3: RGC-TIA

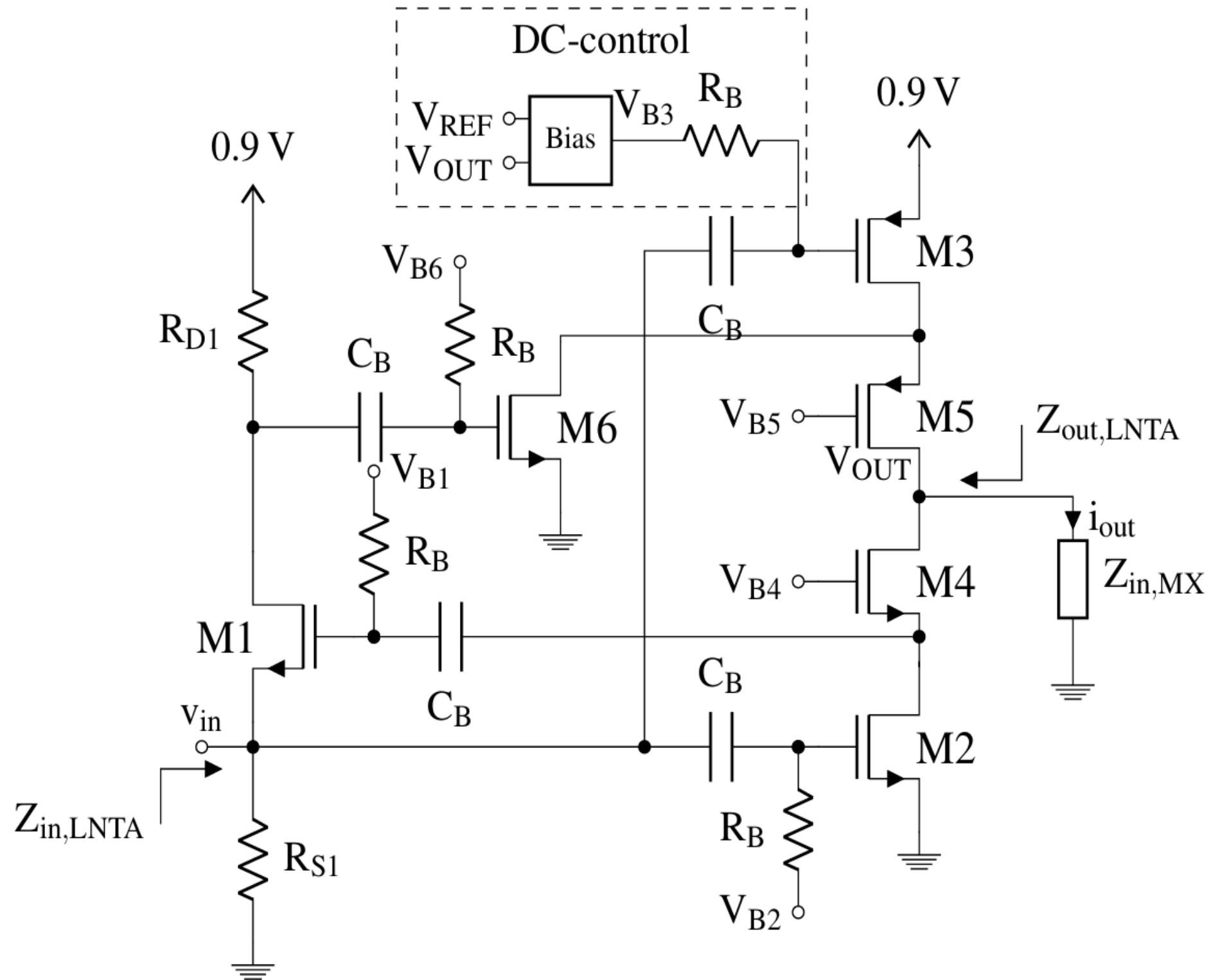
Differential Regulated
Cascode Transimpedance Amplifier



Stage 3: RGC-TIA

Differential Regulated Cascode
Transimpedance Amplifier

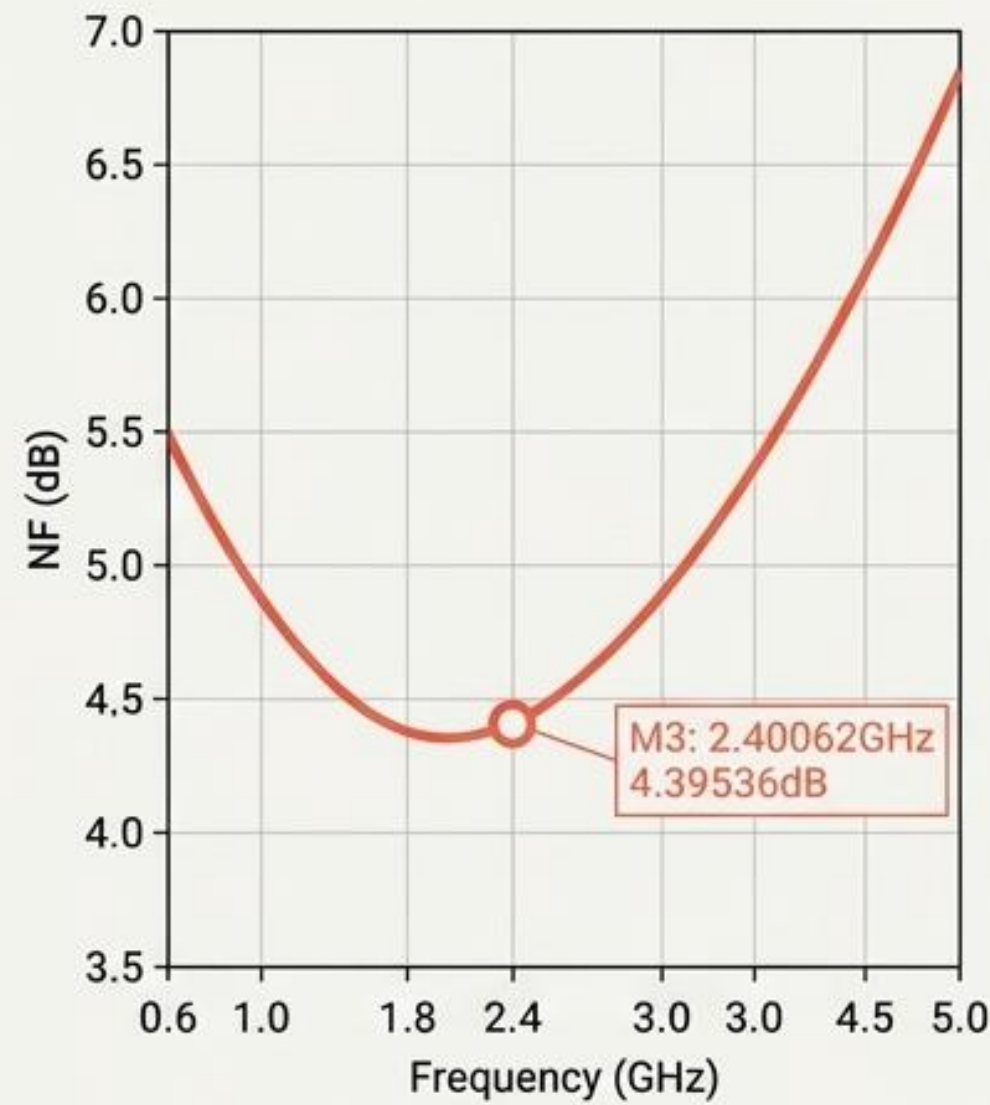
The LNTA: High Transconductance Engine



LNTA Performance Verification

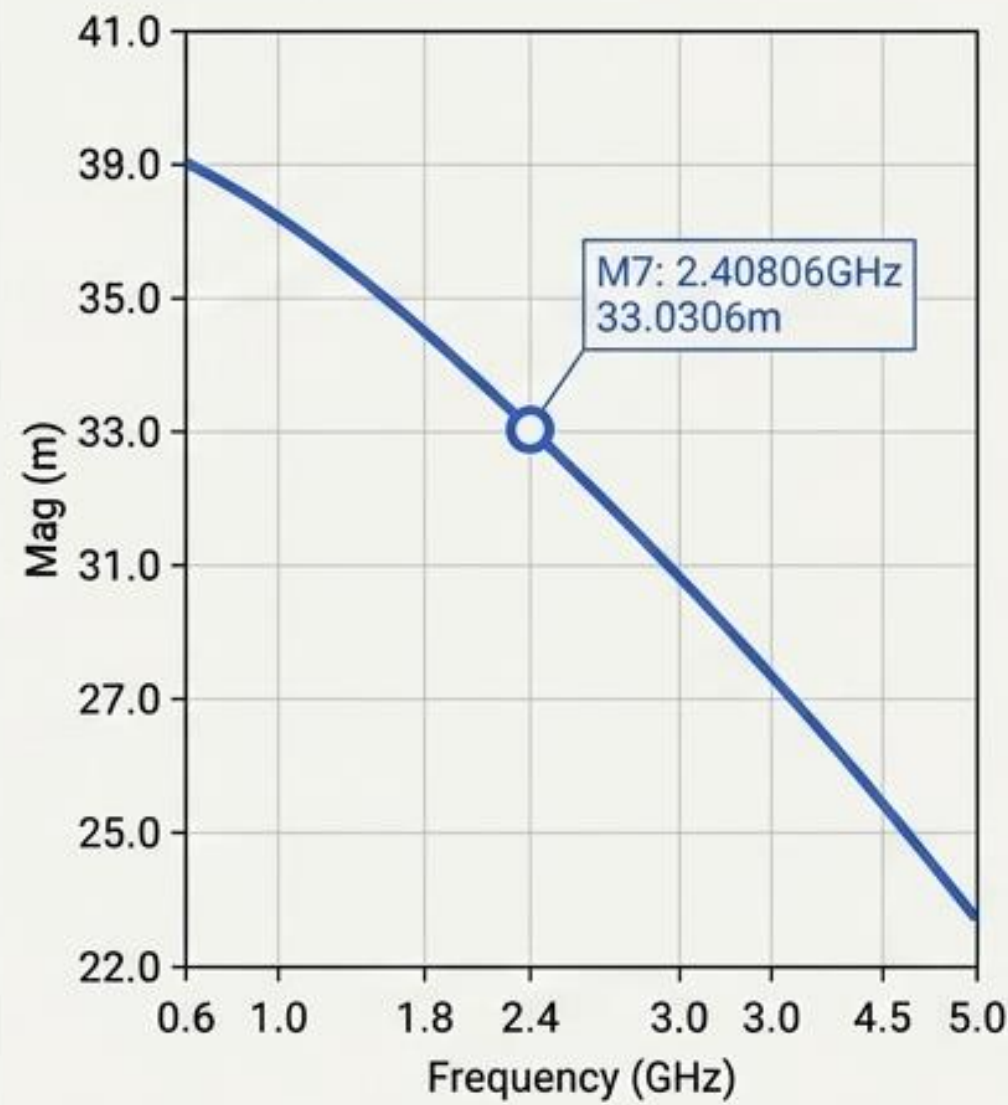
4.39 dB

Minimum Noise Figure



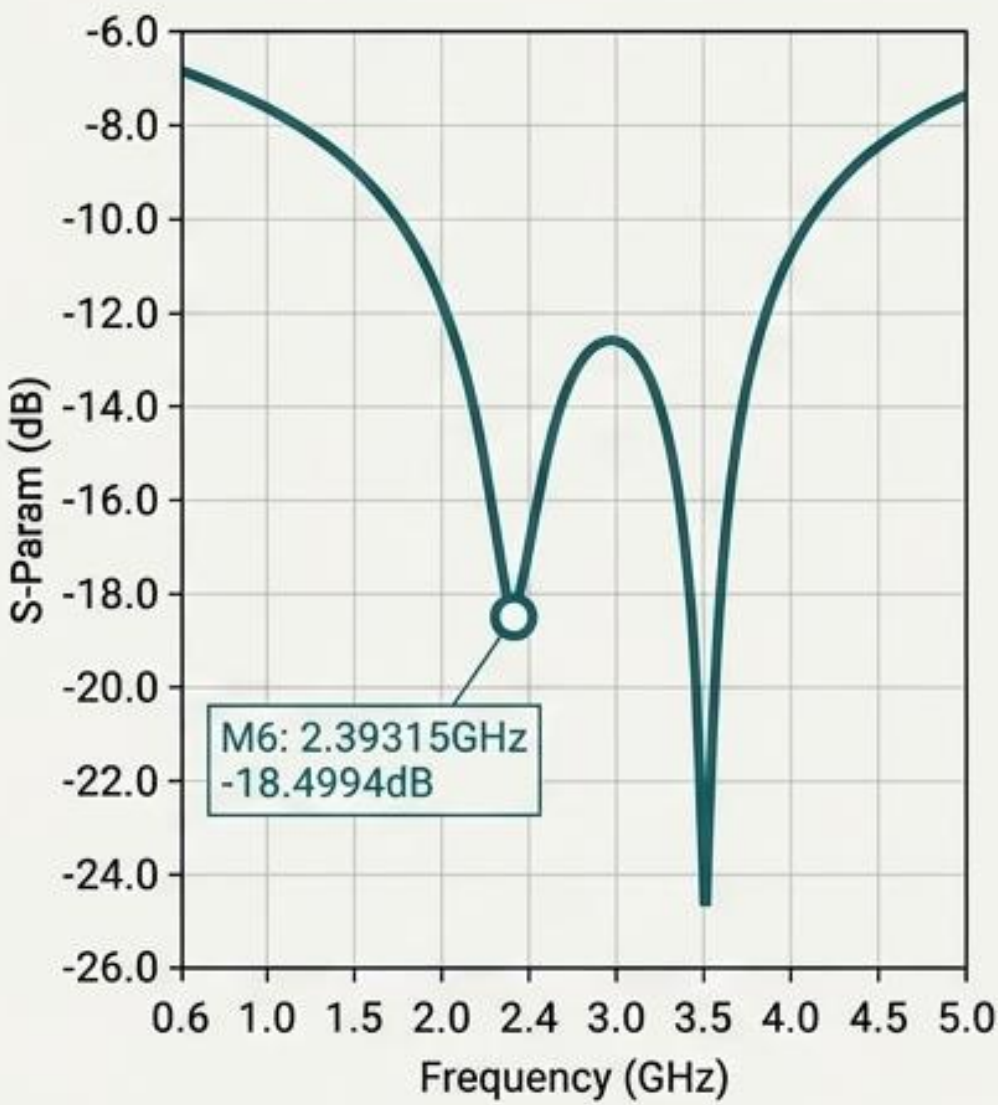
33.03 mS

Transconductance Gain



-18.49 dB

S11 Input Reflection



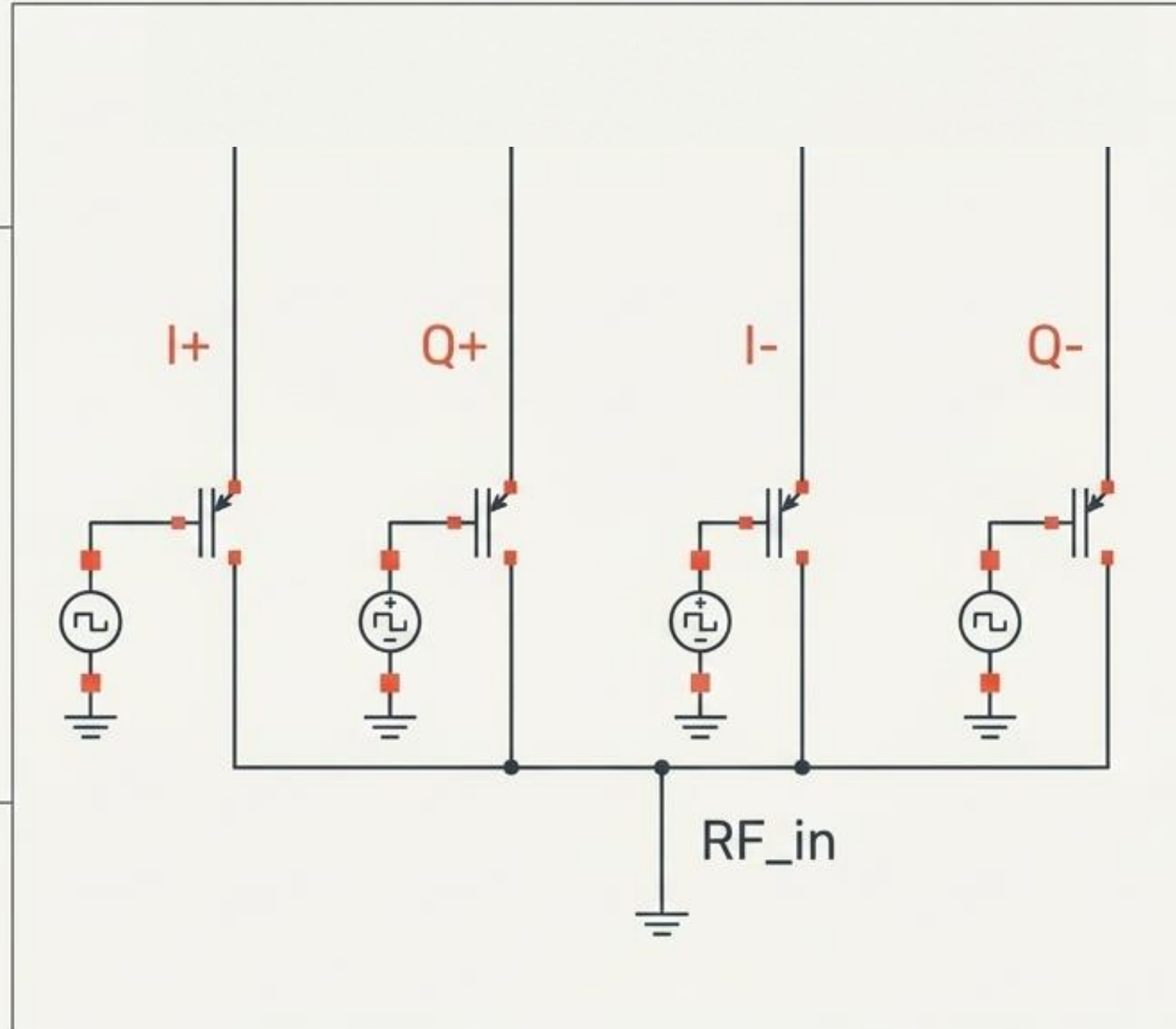
The Mixer: 4-Phase Passive Down-Conversion

Current-Driven Interface

Seamlessly accepts the pure current output from the preceding LNTA without enforcing a voltage swing.

Zero DC Power

The entirely passive switch architecture consumes 0 mW of static DC power, static DC power, preserving the strict 5mW budget.



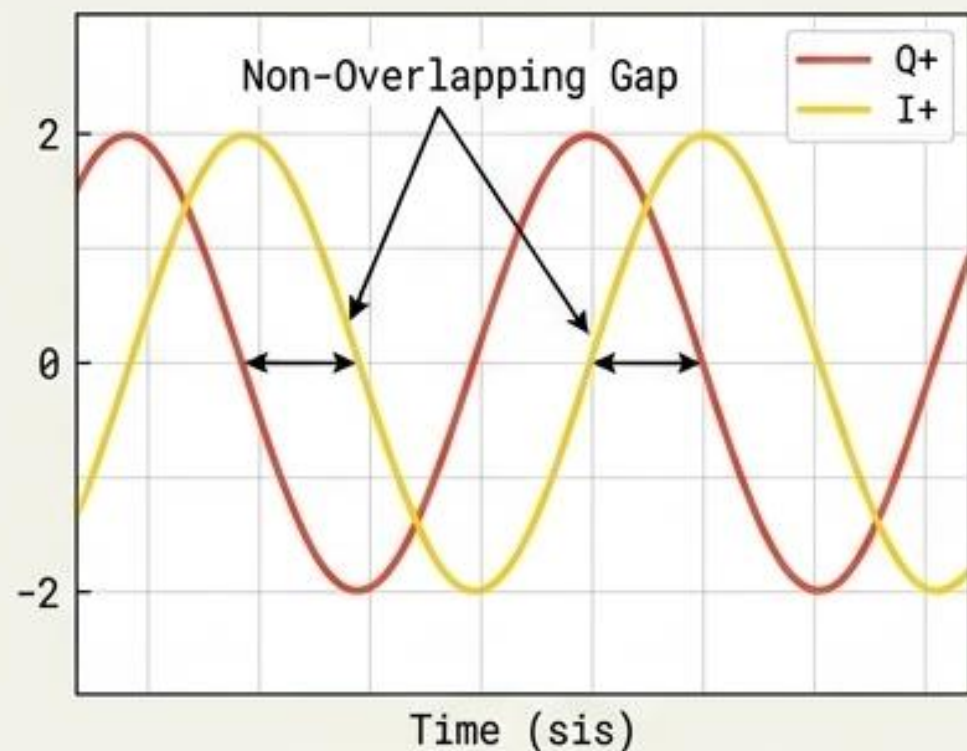
1/f Noise Elimination

Because no DC current flows through the passive switches, flicker noise is fundamentally eliminated—a mandatory trait for modern low-IF receivers.

The Physics of 25% Non-Overlapping Clocks

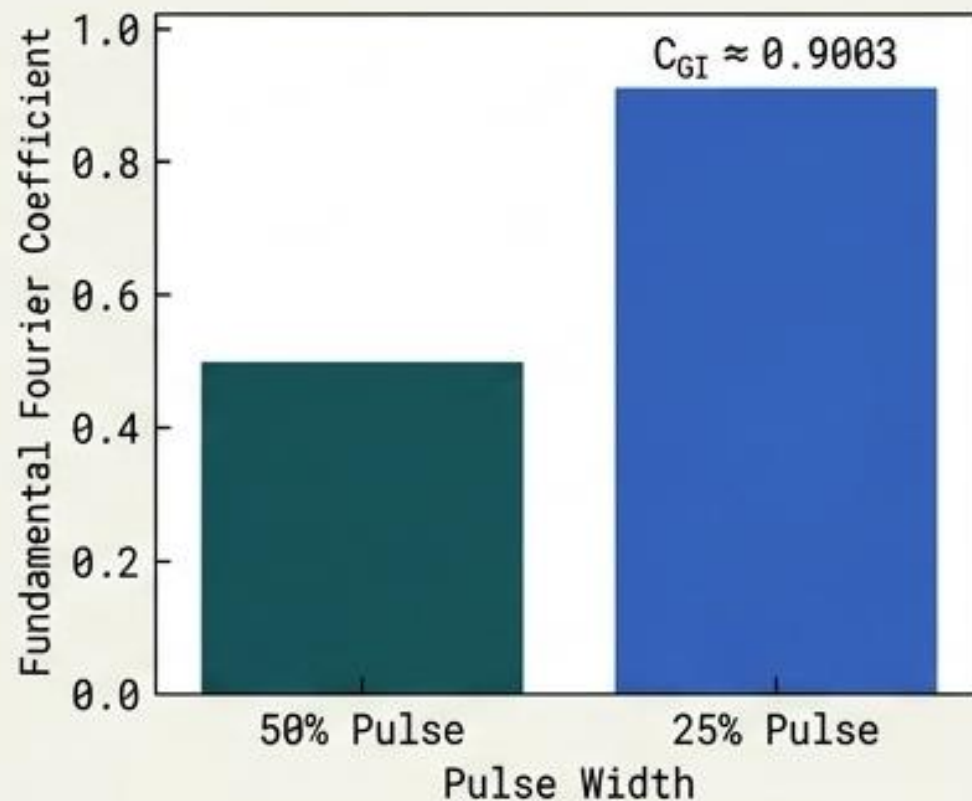
Why 25% duty cycle drastically outperforms standard 50% mixing.

Zero Crosstalk



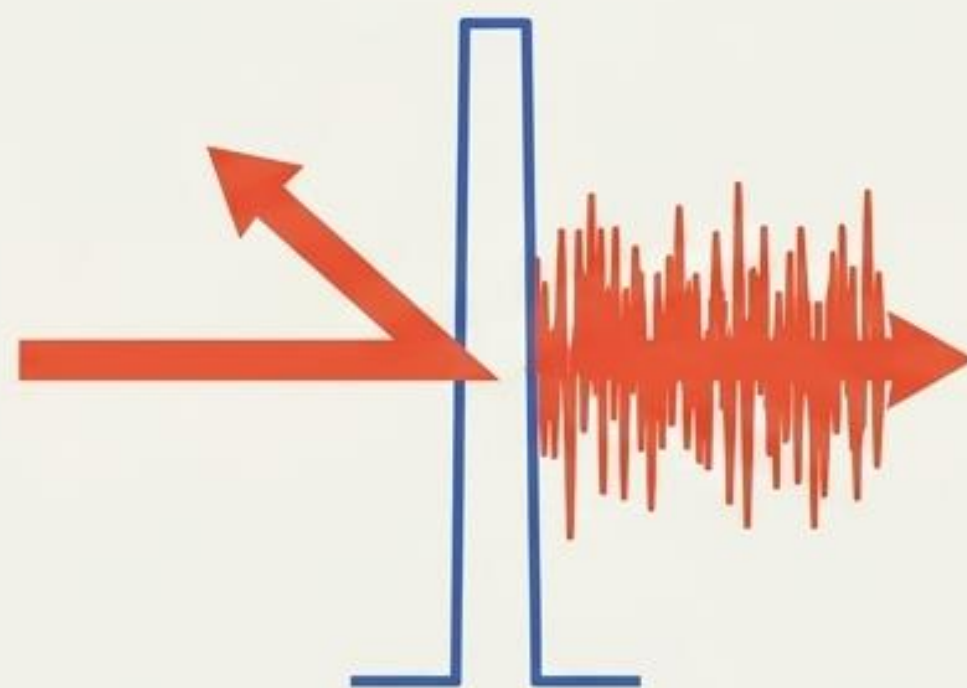
In-Phase and Quadrature channels never conduct simultaneously, entirely eliminating signal leakage.

Improved Conversion Gain



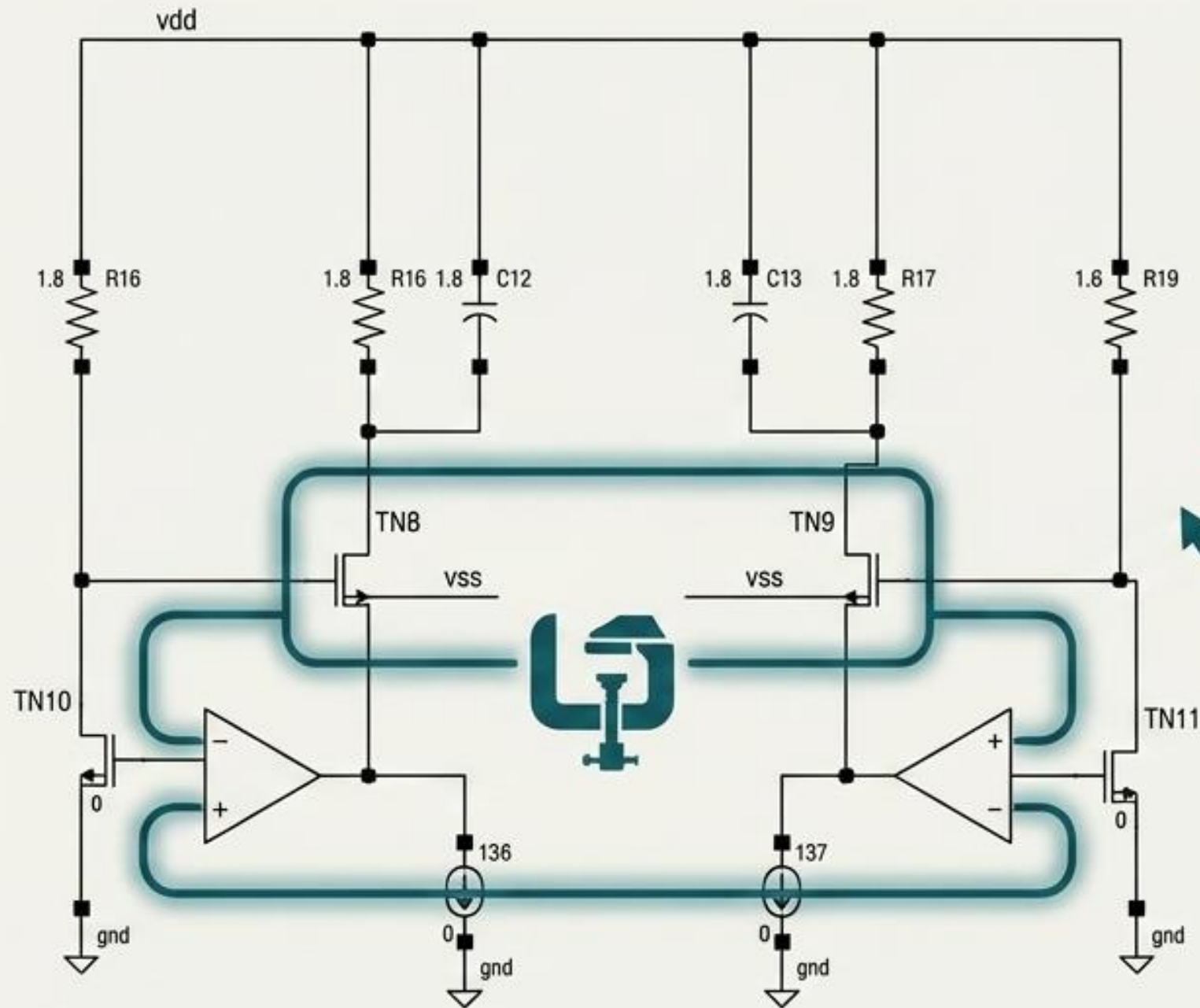
The fundamental Fourier coefficient of a 25% pulse yields higher theoretical conversion gain ($C_{GI} \approx 0.9003$).

Harmonic Rejection



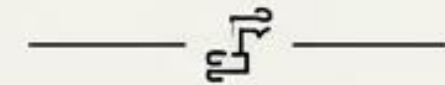
Narrowed pulse width fundamentally alters harmonic folding, rejecting high-frequency thermal noise from down-converting into the baseband.

RGC-TIA: The Virtual Ground Anchor

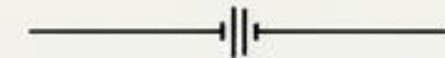


$$Z_{in} \approx \frac{1}{[g_{m,CG}(1 + A_{v,aux})]}$$

The Baseband Challenge: Down-converted 1 MHz IF current must become voltage again, but doing so directly would cause massive voltage swing at the mixer output, destroying linearity.



The RGC Solution: The Regulated Cascode uses a local active feedback loop ($A_{v,aux}$) to rigidly crush input impedance (Z_{in}) toward zero.



The Result: The mixer output is fiercely anchored to a virtual ground.

Baseband Isolation and Output Yield

First-Order Low-Pass Isolation

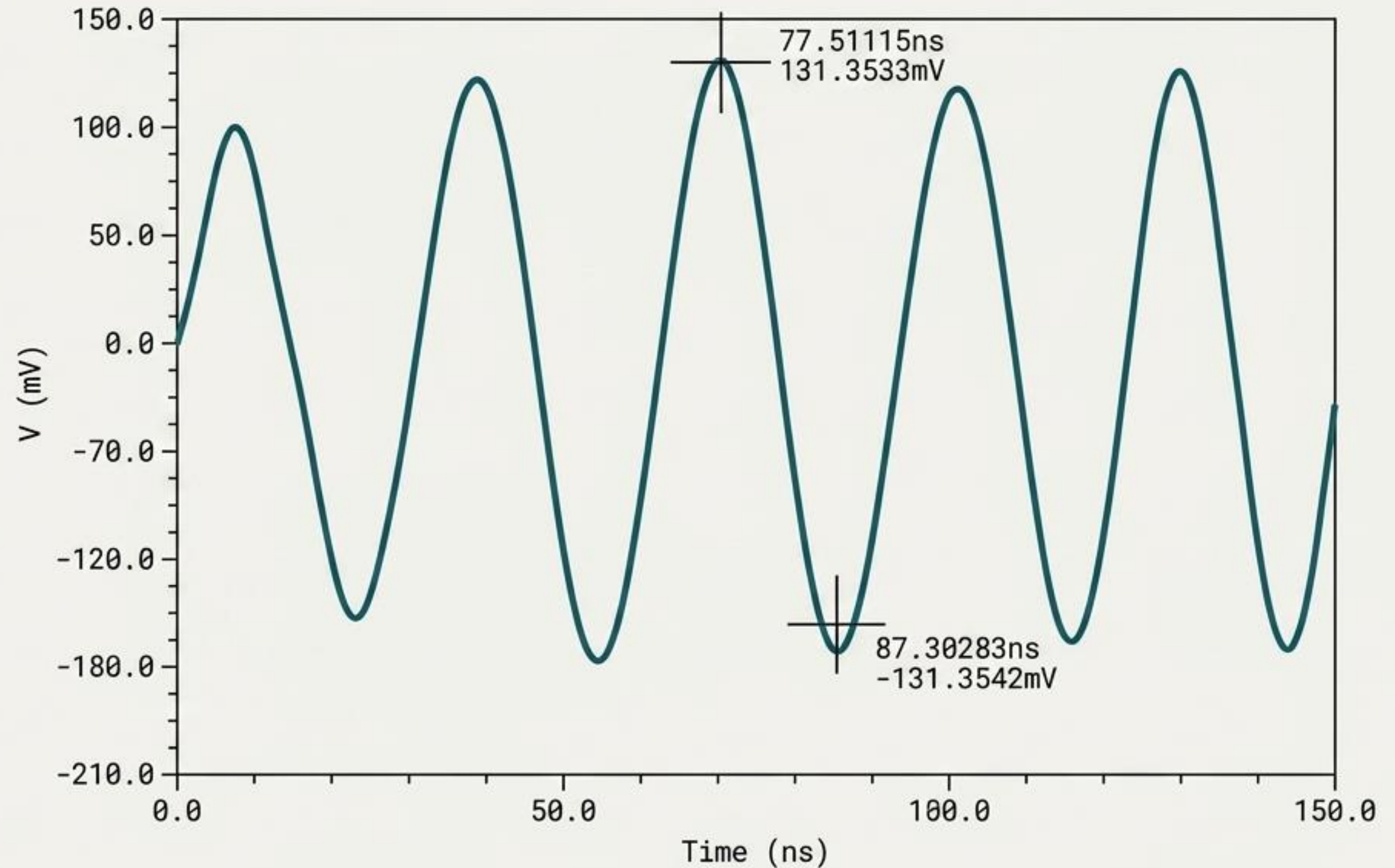
$R_L = 13.9 \text{ k}\Omega$ and $C_L = 4 \text{ pF}$
creates an $f_{3\text{dB}} \approx 2.86 \text{ MHz}$.

High-frequency LO feedthrough
and out-of-band blockers are
systematically destroyed.

Final Signal Integrity

Clean 1 MHz IF Signal
successfully recovered at
the baseband.

Peak Amplitude: 131.35 mV



Total System Performance Synthesis

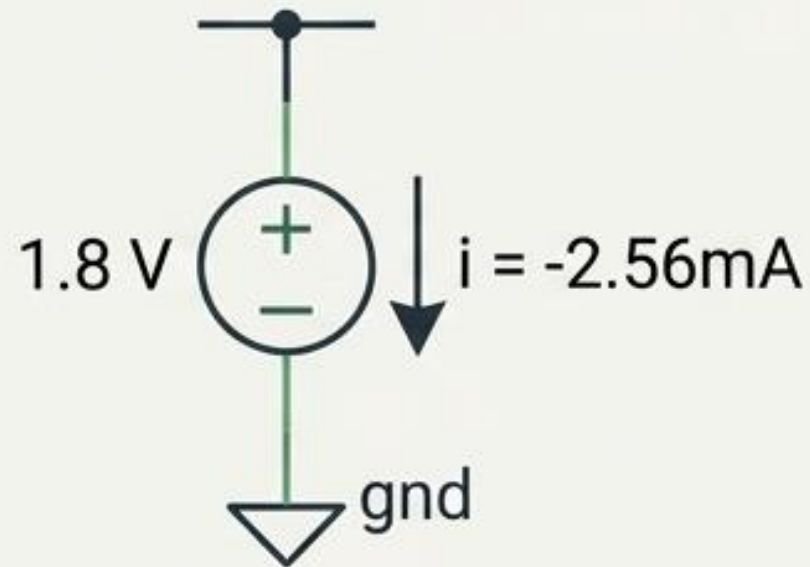
+42.34 dB

System Voltage Conversion Gain

Achieved via massive current translation in the LNTA and rigid impedance anchoring in the TIA.

4.608 mW

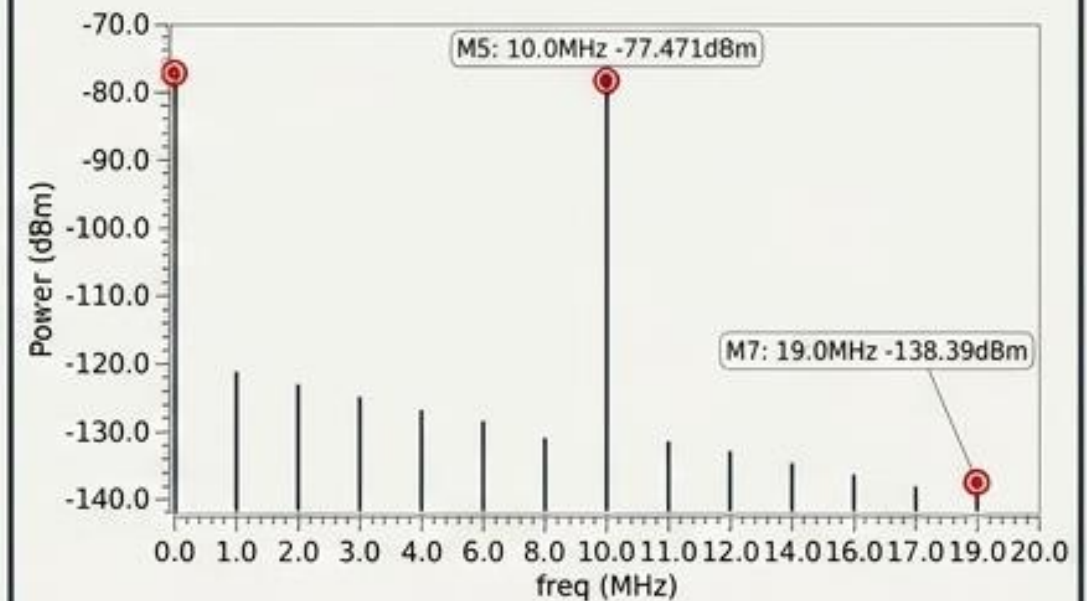
Total Power Dissipation at 1.8V



Easily satisfies the ultra-low power budget for BLE IoT devices.

+10.5 dBm

IIP3 Linearity



Exceeds the > -5 dBm requirement, ensuring extreme resilience to close-in blockers.